VIRTUAL POKER NIGHT

Developer Product & Technical Specification

Core idea: A browser-based virtual poker room where poker is the activity, but the real product is the feeling of sitting around a table with friends. Players see and hear each other live, represented by stylized 3D avatars with their real webcam faces embedded into the avatar heads.

1. Product Vision

This should feel like a beautiful, funky virtual casino rather than a conventional online poker website. Friends enter through a shared link, appear around the same 3D poker table, talk naturally, look around at one another, and physically interact with cards and chips. Poker is the activity, social presence is the differentiator.

Positioning: “Virtual poker night.” The product solves the problem of friends who want to recreate the social atmosphere of playing poker together when they are remote.

2. Locked Product Decisions

Decision	V1 choice
Seating	Fixed seats. No walking/exploration.
Camera	First-person only.
Looking	Smooth mouse-controlled left/right look; player can look down at cards.
Players	Build for 2–6 first, architecture ready for 10.
Avatars	Stylized 3D avatars.
Faces	Player's live webcam face displayed directly on avatar face.
Voice	Live voice by default, with mute/camera controls.
Rooms	Anyone can create a private room and invite friends by URL/code.
Accounts	No account required for V1 unless needed technically.
Visual style	Beautiful, funky, premium casino. Stylized, not photorealistic.
Platform	Desktop browser first, Mac + Windows.
Poker	Texas Hold'em with fake/virtual chips only.

3. Core User Journey

1. Landing: User opens the website and sees Create Room / Join Room.

2. Create: Host creates a room and receives a short shareable URL/code.

3. Invite: Host sends the link to friends.

4. Join: Friend enters a display name, chooses an avatar, and grants camera/mic permissions.

5. Seat: Player is automatically placed into an available fixed seat.

6. Social: Live face and voice become available; player can look around the table.

7. Play: Texas Hold'em begins. Cards and chips are physical 3D objects.

8. Repeat: After each hand, the next hand starts without returning to a lobby.

4. 3D World & Art Direction

The environment should be compact and highly polished. Do not build a large explorable casino for V1. The poker table is the hero asset.

- Premium/funky casino: rich wood, velvet, gold, neon accents, glass and tasteful lighting.
- Stylized low-poly or mid-poly assets to keep browser performance strong.
- Memorable, slightly exaggerated avatar designs and outfits.
- Background casino elements can animate subtly but must not distract from poker or faces.
- No player locomotion in V1.
- No third-person camera in V1.

5. First-Person Camera

- Camera starts at the player's seated eye position.
- Horizontal mouse movement smoothly rotates toward neighboring players.
- Player can look down at their cards/chips.
- Constrain vertical rotation to prevent breaking the scene.
- Use smooth interpolation and subtle head/idle movement if it improves immersion.
- The experience should feel like physically sitting at a table, not controlling a free camera.

6. Avatar + Webcam Face System

The avatar should be stylized. The webcam provides the human identity layer. Avoid realistic full-body human rendering in V1.

- Start with a small library of distinct avatars: cowboy, businessman, gentleman, wizard, alien, shark, etc.
- Avatar consists of head, body, outfit and optional accessory.
- Live webcam feed is cropped/positioned as an oval or face-shaped texture on the avatar's face.
- Background removal/segmentation is optional. A clean crop is sufficient for V1.
- Use lower-resolution streams for distant players and potentially higher quality for the player currently being viewed.
- Add simple idle animations.
- Optional enhancement: microphone activity controls simple mouth-open animation. No AI lip-sync required.

7. Voice & Video

- Use WebRTC infrastructure, preferably LiveKit, rather than building video calling from scratch.
- Camera and microphone are requested on join.
- Voice is open by default after permission.
- Clear camera and mute controls.
- Muted players show a mute icon on/near the avatar chest.
- Show a speaking indicator when a player is talking.
- All players can hear one another in V1. Spatial audio is optional later.
- Do not record or persist audio/video in V1.

8. Poker & Physical Interaction

The poker experience should feel physical rather than like clicking a conventional poker UI.

- Texas Hold'em.
- Server-authoritative deck, hands, turns, bets, pot, community cards, folds and winner.
- 3D cards sit on the table.

• Player can pick up/inspect their cards and put them back face-down.

• Other clients must never receive another player's hidden cards.

• Use fake chips only.

• Prefer physical chip interactions: grab/select denominations and push chips toward the pot.

• Provide Fold, Check, Call and Raise controls as clear UI fallbacks.

• Cards/chips should have satisfying motion and sound effects.

9. Recommended Architecture

Use an authoritative server. Clients send intents; the server validates and broadcasts the resulting state.

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Browser
■■■ React + TypeScript UI
■■■ Three.js / React Three Fiber 3D scene
■■■ LiveKit audio/video
■■■ Game networking

Server
■■■ Room management
■■■ Authoritative poker engine
■■■ Seat/player state
■■■ Game state synchronization
■■■ Action validation

Persistence
■■■ Supabase/Postgres for room metadata and future accounts
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10. Suggested Technology Stack

Layer	Recommendation	Purpose
3D	Three.js + React Three Fiber	Browser 3D rendering
Frontend	React + TypeScript + Vite	UI and room experience
Video/voice	LiveKit / WebRTC	Realtime webcam and microphone
Game networking	Colyseus or Node.js WebSocket	Realtime authoritative game state
Backend	Node.js + TypeScript	Rooms and poker server
Database	Supabase/Postgres	Rooms, guest/player data, optional history
Hosting	Vercel/Cloudflare + server host	Web app and backend deployment
3D assets	Blender + purchased/created assets	Avatars, casino, cards, chips

11. Multiplayer Game-State Rules

• The server owns the deck and all poker state.

• Clients never determine winners.

• Clients never receive hidden cards belonging to other players.

• Validate turn order, bet size, stack size and legal actions on the server.

• Room state should support reconnecting players.

• Use deterministic event/state messages so every client renders the same public table state.

• Do not trust client-supplied balances, cards, seat ownership or game outcomes.

12. Performance Requirements

• Primary target: modern MacBook Air and equivalent Windows laptops in Chrome/Safari/Edge.

• Aim for 60 FPS on capable hardware with graceful degradation.

• Keep the scene compact and optimize draw calls, textures, lighting and shadows.

• Use compressed textures and reasonable texture resolutions.

• Avoid expensive real-time global illumination and excessive dynamic shadows.

• Reduce webcam stream resolution for players who are not the visual focus.

• Add a quality setting or automatic fallback if GPU performance is weak.

• Initial target is 6 players; architecture should support 10.

13. MVP Must-Haves

• Create private room and share URL/code.

• Join room as a guest with display name.

• Avatar selection.

• Camera/microphone permissions.

• Live webcam face mapped to avatar.

• Live voice and mute/camera controls.

• Fixed seats.

• First-person mouse-look.

• 2–6 player Texas Hold'em.

• Private hole cards.

• Fold/check/call/raise.

• Virtual chips and pot.

• Community cards and showdown.

• Server-authoritative poker state.

• Play multiple hands in the same room.

• Basic reconnect handling.

14. Explicitly Out of Scope for V1

• iOS/Android native app.

• Player walking or open-world exploration.

• Real-money wagering, deposits or withdrawals.

• Public matchmaking.

• Friend/account system unless required.

• Tournaments.

• Cosmetics marketplace.

• AI players.

• Multiple casino locations.

• Full-body motion capture.

• AI lip-sync.

• Advanced spatial audio.

• Complex NPCs or large casino environments.

15. Development Phases

Phase	Goal	Definition of done
0	Technical spike	Two browser tabs connect and share realtime camera/mic plus basic state.
1	Social prototype	Two 3D avatars with live faces can look at each other and talk.
2	Poker prototype	Two players can complete a full Hold'em hand.
3	Multiplayer	3–6 players can join, sit, talk and play.
4	Physical interaction	Cards/chips have polished 3D interactions and sounds.
5	Visual polish	Funky premium casino, lighting, animations and UI.
6	Reliability	Permissions, reconnects, edge cases, browser compatibility and performance.
7	Private beta	Test with real friend groups and iterate on the social experience.

16. Privacy & Security

• Use HTTPS/WSS in production.

• Use secure, difficult-to-guess room tokens.

• Do not record or persist webcam, microphone or voice data in V1.

• Camera/microphone permissions must be explicit and reversible.

• Poker cards are private server data.

• Rate-limit room creation and abusive requests.

• Use fake chips only. Do not implement real-money gambling in V1.

17. Estimated Initial Cost

If built by the founder/developer: roughly $20–$300 upfront is a reasonable planning range for a private prototype, excluding development time. Open-source frontend/3D technologies are free, and realtime/database/hosting services can begin on free or low-cost tiers depending on usage. A domain and 3D assets are the most likely cash expenses.

If outsourced: roughly $10,000–$30,000+ is a reasonable planning range for a polished MVP, depending heavily on engineering, 3D art, animation and polish. These are planning estimates, not quotes.

18. Browser & Device Strategy

Do not build a native iOS app for V1. The intended experience is well suited to desktop browsers because it relies on webcam, microphone, mouse look and a larger screen. Modern MacBook Air hardware should be a realistic target for a well-optimized Three.js scene with a small number of stylized avatars. The application should also be tested on equivalent Windows laptops.

19. Future Expansion

If poker night proves that the social experience is compelling, the same virtual room can become a broader private hangout platform.

• Blackjack

• Uno-style card games

• Trivia

• Board games

• Party games

• Persistent friend groups

• Avatar customization

• Different casino rooms/themes

• Spectator mode

• Virtual-chip tournaments

20. Developer Brief, Copy/Paste

Build a desktop-browser multiplayer Texas Hold'em experience centered around social presence. Players join private rooms through shareable links. Each player occupies a fixed seat at a stylized, premium/funky 3D casino poker table. The camera is first-person and controlled by mouse movement. Players see each other as stylized 3D avatars, with each avatar's face replaced or overlaid by the player's live webcam feed. All players can talk through live microphone audio and can mute themselves. Poker cards and chips should behave as physical 3D objects. The server must authoritatively manage all poker state and private card information. The first release should support 2–6 players, while the architecture should be extendable to 10. It must run in a modern desktop browser on Mac and Windows, with no native iOS app required for V1. The priority is not building a huge casino. The priority is reproducing the emotional moment of sitting around a poker table with friends who are physically somewhere else.

21. Definition of Done

• A host creates a private room and shares a link.

• At least six players can join.

• Each player is assigned a fixed seat.

• Players see and hear one another live.

• Each real webcam face appears on the selected 3D avatar.

• Mouse-look feels smooth and seated.

• Players can look down and interact with their cards.

• A complete Hold'em hand can be played.

• Hidden cards remain private.

• Game state is server-authoritative.

• Players can immediately start another hand.

• The casino feels visually cohesive and polished.

• The game is usable on a modern MacBook Air without a native app.

Product principle: Every technical and design decision should serve one outcome: “Holy shit, it feels like we're actually sitting at the table together.”